

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B. Tech CSE**

Semester: VI

Name of the Course: Network and System Security

Course Code: TCS619

Time: 90 Minutes

Maximum Marks: 50

Note:

- i. Answer all the questions by choosing any one of the sub questions.
- ii. Each questions carries 10 marks.

Q1		(10 Marks)	CO1
(a)	Discuss the role of encryption, digital signatures, and authentication exchange as security mechanisms in X.800. How do these mechanisms contribute to the CIA triad in secure communication networks?		
	OR		
(b)	What is a computationally secure scheme in cryptographic systems? Explain how public-key cryptography (i.e., RSA) ensures computational security in secure communications.		
Q2		(10 Marks)	CO2
(a)	Given the encryption function of Affine Cipher: $C=(5P+8) \bmod 26$ where P is the plaintext letter ($A = 0, B = 1, \dots, Z = 25$). Encrypt the plaintext "WELCOMETOGEU" using this function. Analyze the strength and vulnerabilities of the Affine Cipher in terms of cryptographic security.		
	OR		
(b)	Demonstrate the scenario of the man-in-the-middle attack (MITM) using a figure.		
Q3		(10 Marks)	CO3
(a)	Describe the following general security requirements. Data freshness, authentication and non-repudiation.		
	OR		
(b)	Demonstrate the working of the Cipher Block Chaining (CBC) mode of encryption and decryption with the help of block diagrams. How does the use of an Initialization Vector (IV) improve security?		
Q4		(10 Marks)	CO5
(a)	Explain the step-by-step process of key generation, signature generation, and signature verification in the RSA digital signature algorithm. How does the security of the algorithm rely on the computational difficulty of the integer factorization problem?		
	OR		

(b)	Demonstrate the working mechanism of Kerberos. What are the roles of the following entities in Kerberos, i.e., Key Distribution Center (KDC), Ticket Granting Server (TGS), and Authentication Server (AS)?	
Q5	(10 Marks)	CO2
(a)	Describe the concept of authentication tied to ciphertext in cryptographic security using a figure. Discuss the security aspects of this mechanism.	
	OR	
(b)	Compare symmetric-key cryptosystems and asymmetric-key cryptosystems in terms of key management, security, and computational efficiency.	